

House Price Prediction Project

Summary of Findings

Project Overview

The objective of this project was to develop a machine learning model capable of predicting house prices using property features such as area, number of bedrooms, bathrooms, parking availability, and other amenities. Data preprocessing, exploratory analysis, and model evaluation were performed to identify the most influential factors affecting house prices.

Key Findings

- Missing values were handled using median imputation, and duplicate records were removed to improve data quality.
- Categorical variables were converted into numerical form using one-hot encoding to make them suitable for machine learning algorithms.
- Other important factors influencing house prices include:
 - Number of bathrooms
 - Number of bedrooms
 - Parking availability
 - Air conditioning
 - Additional amenities
- Two regression models were trained and evaluated:
 - Linear Regression
 - Random Forest Regressor
- The **Random Forest Regressor outperformed Linear Regression**, indicating its ability to capture complex and non-linear relationships within the housing dataset.
- Feature importance analysis confirmed that property size (area) is the most significant factor in determining house prices.
- An interesting observation was that houses with similar sizes often had different prices due to variations in amenities and location-related characteristics.

Conclusion

The developed machine learning model successfully predicts house prices with good accuracy. The results demonstrate that house area, room count, and available facilities play a major role in determining property value. Random Forest proved to be the most effective model for this dataset and can be used as a reliable tool for house price estimation.